

COMP 3000 (WINTER 2023) OPERATING SYSTEMS

ASSIGNMENT 4

Please submit the answers in Brightspace by the due time indicated in the submission entry. There are 10 points (weight: 0.25) in part 2 and part 1 is a short quiz in Brightspace worth 10 points (weight: 0.25).

Submit your answers as a **gzipped tarball** "username-comp3000-assign4.tar.gz" (where username is your MyCarletonOne username). Do NOT just submit a file of another format (e.g., txt or docx) by renaming it to .tar.gz. Unlike tutorials, assignments are graded for the correctness of the answers.

The tarball you submit **must** contain the following (applicable to all assignments):

1. A **plaintext** file containing your solutions to all questions, including explanations.
2. A README.txt file listing the contents of your submission as well as any information the TAs should know when grading your assignment. Without this file, grading will be based on the TA's understanding.
3. For each question, where applicable, a C file for your modified version of the original file. This should include all required changes for that question. This way, you can avoid something wrong with one question affecting another question.
4. Diff files showing the modifications, by comparing each submitted C file above and the original: for example, `diff -c 3000original.c 3000modified.c > 3000.diff`. Avoid moving around or changing existing code (unless necessary) which may be distracting.

For this assignment, you only need to submit code for question 1 in part 2.

You can use this command to create the tarball: `tar zcvf username-comp3000-assign4.tar.gz your_assignment_directory`. ****Don't forget to include your plaintext file!!****

No other formats will be accepted. Submitting in another format will likely result in your assignment not being graded and you receiving no marks for this assignment. In particular, do not submit an MS Word, OpenOffice, or PDF file as your answers document!

As a student learning operating system, you are expected to be able to handle file transfer properly using necessary commands such as `ssh`, `scp` or `rsync`. Otherwise (which happens), **please set aside time by completing your assignment enough long before the due time**, then you may ask a TA, the instructor, or another student for any file transfer issue. Then, you can keep working on it and **override your submission** by uploading a new tarball before it is due.

For your convenience, [here are a few commands](#) (among many others) to download a file from your Openstack VM. You need to run the command on your PC, not your Openstack VM.

Empty or corrupted tarballs may be given a grade of zero, so please **double check your submission by downloading and extracting it**.

Don't forget to include what outside resources you used to complete each of your answers, including other students, and web resources. You do not need to list help from the instructor, TA, or information found in the textbook or man pages.

Use of any outside resources verbatim as your answer (like copy-paste or quotation) is not allowed, and will be treated as unauthorized collaboration (and reported as plagiarism).

Please do **NOT** post assignment solutions on MS Teams or Brightspace (or other platforms not used in the course such as [Discord](#)) or it will be penalized. Moreover, [posting the assignment questions to any external forums/websites is NOT permitted and will also be penalized/reported](#).

Questions – part 1 [10]

This part is a short quiz in Brightspace with only multiple-choice questions and will be posted in a few days. Look for “Assignment 4 – Part 1” under Quizzes.

Questions – part 2 [10]

1. [6] Based on `ones.c` in Tutorial 7, try to change its behavior so that it outputs what you tell it to and can be “muted” (becoming a `/dev/null`). The new `/dev/ones` should behave as follows:
 - At the beginning it still outputs ‘1’s like before, as no data has been written.
 - It takes one byte each time you write to it, e.g., `echo -n “h” > /dev/ones` will have the same effect as `echo -n “hello” > /dev/ones`
 - Once written to, `/dev/ones` should start outputting that byte it has just received. In the example above, it will always output “hhhhhhhhh...” (one ‘h’ at a time, like one ‘1’ at a time originally), until you write something else.
 - Whenever it receives the letter ‘n’, it will be muted, meaning: subsequent reads will just get EOF, exactly like reading from `/dev/null`.
 - Later, when you write something else (other than ‘n’), it will resume outputting the new received character.
2. In Tutorial 7, you should have used the command `insmod` a lot to insert a kernel module. Now, let’s think about how this command achieves its purpose: find out what specific system call it makes to insert a kernel module. Hint: you can use `strace`.
 - a. [1] Write down the name of the system call (it has to be one with the correct spelling).
 - b. [1] Why was `mmap()`/`munmap()` needed when inserting a kernel module? (be specific, based on question a above.) You can make use of the man pages.
3. [2] Try the following command:

```
$ cat < /dev/tcp/127.0.0.1/22
```

(you’ll need to explain what makes this work)

You should see something like (the version numbers may vary):
SSH-2.0-OpenSSH_7.6p1 Ubuntu-4ubuntu0.5

This means your local SSH daemon is working (`127.0.0.1` refers to localhost and `sshd` listens on port 22). You can also replace the file with `/dev/tcp/access.scs.carleton.ca/22` to test the connectivity to the gateway machine where your files are backed up.

But if you try to check that file or directory using any other command:

```
$ ls /dev/tcp
ls: cannot access '/dev/tcp': No such file or directory
$ ls /dev/tcp/127.0.0.1/22
ls: cannot access '/dev/tcp/127.0.0.1/22': No such file or directory
$ cat /dev/tcp/127.0.0.1/22
cat: /dev/tcp/127.0.0.1/22: No such file or directory
```

Based on the OS knowledge you have learned so far, what could be happening? Be specific about who (which program) could be supporting this functionality, with some explanation.

Your answer can be either the accurate technical description or your speculation that technically makes sense.

Do not just say something like it tries to establish a connection to the port of that host using TCP (which is obvious). Also, the OS/kernel is not aware of this the user-space command’s semantics so that’s not the answer.